Multilingual Document Processing Architecture

1. Introduction to Digital Text Normalization

Digital document preservation requires accurate conversion from vector formats into semantic HTML. Traditional PDF viewers rely on canvas element rendering or pre-rasterized tiles, losing real text semantics. By generating clean DOM text nodes, screen readers and editors can freely select, search, and annotate text.

2. Core Architectural Principles

- Semantic HTML elements instead of absolute position divs
- Durable text quote and Unicode position anchoring for annotations
- Multi-language OCR integration supporting Indic and RTL scripts
- Offline-first self-contained ZIP bundles

Mathematical Formulation

E = mc^2 and sum(x_i) = X

3. Performance Benchmarks

Conversion benchmarks across varying document complexities demonstrate bounded memory utilization.